

Education Background

École de technologie supérieure (ETS)
Ph.D in Electrical Engineering
University of Southern California (USC)
M.S in Electrical Engineering
Shenyang University of Technology
B.S in Electronic Information Engineering

Montreal, QC, Canada
08/2019-Present
Los Angeles, CA, U.S.A
08/2017-05/2019
Shenyang, Liaoning, China
09/2013-07/2017

TECHNICAL SKILLS

Inference Serving / ML Systems: LLM Serving, TTFT, TPOT, Throughput, P95 Latency, Prefill / Decode Analysis, KV Cache, Concurrency Benchmarking, Latency Profiling, Performance Analysis, PyTorch, Transformers, GPU Benchmarking
Distributed Training / Communication: NCCL, Collective Communication (AllReduce, ReduceScatter, AllGather), Data Parallelism (DP), Tensor Parallelism (TP), Pipeline Parallelism (PP), Multi-GPU Benchmarking, Communication Profiling, Synchronization Optimization
LLM Training / Systems Optimization: GPT-Neo-1.3B Training, Memory Optimization, Gradient/Optimizer/Parameter Sharding, Distributed Throughput Tuning, Multi-GPU Training, Training Stability Optimization
GPU Programming: CUDA C++, Triton, Nsight Compute, Warp-level primitives
Performance Optimization: Roofline analysis, memory bandwidth modeling, kernel fusion
ML Systems: FlashAttention, KV cache layout, IO-aware optimization
System Modeling: Multi-agent RL, distributed resource scheduling
Networking Background: 5G, PDMA, DRL (applied to resource optimization)

Related Experience

[LLM Quantization Benchmark & Inference Optimization Project](#)

- Built an **LLM quantization benchmark pipeline** to compare **FP16 inference with FP8, INT8, AWQ, and GPTQ** models under consistent prompt, output-length, batch/concurrency, and hardware settings.
- Measured **TTFT, TPOT, throughput, end-to-end latency, p95 latency, GPU memory footprint**, and **generation consistency** under fixed workload settings.
- Analyzed how reduced **weight/activation precision** affects prefill/decode latency, KV-cache memory pressure, and serving throughput.
- Identified scenarios where quantized inference improved serving efficiency, as well as cases where dequantization overhead, unsupported kernels, or quality degradation reduced the practical benefit.
- Generated deployment-oriented analysis to guide quantization choices for memory-constrained and latency-sensitive LLM serving workloads.

[LLM Serving Benchmark & Inference Systems Analysis Project](#)

Designed and benchmarked **LLM inference workloads** to analyze **serving performance, prefill/decode behavior**, and **concurrency trade-offs** on GPU:

- Built a lightweight **LLM serving benchmark harness** on **T4 GPU** to measure core inference metrics, including **TTFT, TPOT, end-to-end latency, throughput**, and **p95 latency**, across controlled prompt and output settings.
- Implemented single-request experiments to characterize **prefill vs. decode** behavior and analyzed how **prompt length** impacts **TTFT** while **output length** affects **TPOT** and overall generation latency.
- Developed controlled benchmarking sweeps over prompt length and output length, and generated performance plots such as **TTFT vs. prompt length** and **TPOT vs. output length** to identify distinct bottlenecks between **compute-heavy prefill** and **memory-sensitive decode**.
- Extended the benchmark to **multi-request concurrency experiments** under concurrency levels from **1 to 16**, measuring **throughput scaling, average latency**, and **tail latency** to study **latency-throughput trade-offs** in inference serving workloads.
- Produced **performance analysis reports** with benchmark curves and system insights on why **throughput saturates** under higher concurrency, why **latency degrades** with request contention, and when **prefill/decode bottlenecks** dominate serving performance.

[Distributed Training Systems & NCCL Parallelism Project](#)

Designed and optimized distributed training experiments for large-model parallelism with communication-aware benchmarking:

- Built and benchmarked distributed communication workflows on dual-GPU setup using **NCCL collectives**, including **AllReduce, ReduceScatter**, and **AllGather**, and analyzed bandwidth/latency behavior across message sizes.

- Implemented and compared core large-scale training parallelism strategies, including **Data Parallelism (DP)**, **Tensor Parallelism (TP)**, and **Pipeline Parallelism (PP)**, with hands-on experiments inspired by Megatron-style TP and GPipe-style PP.
- Developed distributed benchmarks to measure communication overhead, scalability, and throughput under different parallel execution patterns and workload granularities.
- Optimized NCCL communication efficiency by improving message granularity, reducing unnecessary synchronization, and refining communication patterns, achieving **25–30% communication overhead reduction** in the project benchmark.
- Produced performance analysis reports with bandwidth curves, latency trends, and parallelism trade-off summaries to explain when **TP/PP/DP** are beneficial under different system constraints.

[Distributed Training Optimization Project \(FSDP / ZeRO / Activation Checkpointing\)](#)

Designed and optimized large-scale distributed training workflows for billion-parameter language models using PyTorch and NCCL:

- Implemented **FSDP full-shard** training for **GPT-Neo-1.3B** to fit billion-parameter training workloads under constrained multi-GPU memory.
- Analyzed **ZeRO-1/2/3** sharding strategies and mapped their impact on parameter, gradient, and optimizer-state memory footprint.
- Integrated **activation checkpointing** to reduce peak activation memory and support larger effective batch sizes.
- Tuned **microbatch size, gradient accumulation, and sharding** configurations to balance memory usage, GPU utilization, and training stability.
- Profiled **NCCL collectives** and synchronization points to identify communication bottlenecks in distributed training.
- Observed up to **2× training throughput** improvement in controlled experiments after tuning microbatching, checkpointing, and sharding configurations.

[CUDA Kernel Optimization & PyTorch Extension Project](#)

Designed and optimized high-performance CUDA kernels for ML workloads with hardware-aware memory tuning:

- Implemented **Matmul, LayerNorm (fwd/bwd), and numerically stable Softmax** in CUDA C++
- Applied **block tiling, shared memory reuse, loop unrolling, and register-level optimization**
- Optimized kernel occupancy and reduced warp stall by tuning block size and register usage.
- Achieved up to **2–3× speedup over PyTorch** eager baselines on selected memory-bound operators such as LayerNorm and Softmax, and **~200× speedup over CPU baselines** in controlled benchmarks.
- Integrated kernels as **PyTorch C++/CUDA extensions with full autograd support**

[IO-Aware Attention System \(Triton + FlashAttention\)](#)

Designed and implemented IO-aware attention kernels to reduce global memory traffic:

- Modeled attention compute complexity ($O(n^2d)$) vs memory traffic ($O(n^2)$) and identified IO-bound bottleneck
- Built naive Triton attention baseline ($QK^T \rightarrow \text{Softmax} \rightarrow V$) with mask support
- Implemented mini **FlashAttention-style kernel** with:
 - Tiled Q/K loading
 - Online softmax (running max + running sum)
 - SRAM reuse and kernel fusion
- Reduced global memory access through tiling strategy
- Achieved **~2× speedup vs naive attention implementation with improved global memory efficiency and reduced DRAM traffic**.

[System-Level Optimization with Multi-Agent RL \(5G / Resource Scheduling\)](#)

- Modeled multi-cell interference and resource allocation as a multi-agent optimization problem
- Designed MADDPG / Actor-Critic frameworks for distributed power scheduling
- Built simulation environments for large-scale communication system optimization
- Focused on system-level efficiency, scalability, and coordination under partial observability

[Kaggle RSNA Screening Mammography Breast Cancer Detection](#) (Rank 94/1687, Top 6%)

- Built DALI-based high-throughput pipeline
- Improved classification robustness via region-focused cropping
- Deployed ConvNeXtV2 & EfficientNet for robust medical image classification

[Kaggle Google - Fast or Slow? Predict AI Model Runtime](#) (Ranking: 30/616, Top 5%)

- Designed GCN-based performance predictor with graph-structured embeddings

Kaggle ICR - Identifying Age-Related Conditions (Rank 407/7327, Top 6%)

- Implemented LightGBM/CatBoost ensemble with Optuna hyperparameter tuning

ITU AI/ML for 5G-Energy Consumption Modelling by ITU(Rank 19/221, Top 9%)

- Built time-series energy prediction pipeline with feature engineering & residual boosting

Previous Employment

Research Assistant(full time)	
ETS – Montreal, Quebec, Canada – Sychromedia	09/2019- Present
• Designed communication-aware optimization frameworks for UAV-assisted 5G and LEO satellite systems, integrating multi-agent DRL for distributed scheduling under dynamic topology.	
Mitacs Research Intern	
Ciena – Montreal, Quebec, Canada – Self-Optimizing Fabric (SOF) project	09/2019- 08/2022
• Delivered progress updates for the SoF project.	• Presented research at top international telecom conferences.
• Explored ML deployment challenges on cache	
Mitacs Research Intern	
Vmware – Montreal, Quebec, Canada	08/2022 – 04/2024
• Published research in peer-reviewed journals and conferences.	• Presented research at top international telecom conferences.
• Developed Reinforcement Learning algorithms	
AICT-Centech Research Intern	
Dromii – Montreal, Quebec, Canada	09/2025 – 12/2025
• Developed CV Algorithm for pothole detection and classification	• Developed CV augmentation algorithms.

Award

• Most creative solution of ITU AI/ML in 5G challenge	12/2023	• Mitacs Accelerate Ph.D intern fellowship (4 semesters)	2023 - 2024
• Excellent Graduate Design for top 3%	07/2017	• Mitacs Accelerate Ph.D fellowship (8 semesters)	2019-2022
• Excellent Undergraduate Academic Scholarship (class two once and class three twice)			2013-2016
• National First prize National College Students Connected Smarter System Innovation Competition			07.2016

Publications

Ramachandran, G. S., Wright, K. L., **Zheng, L.**, Navaney, P., Naveed, M., Krishnamachari, B., & Dhaliwal, J. (2019, May). Trinity: A byzantine fault-tolerant distributed publish-subscribe system with immutable blockchain-based persistence. In *2019 IEEE International Conference on Blockchain and Cryptocurrency (ICBC)* (pp. 227-235). IEEE.

Zheng, Licheng, Kim Khoa Nguyen, and Mohamed Cheriet. "NSATC: An interference aware framework for multi-cell NOMA TUAV airborne provisioning." *2022 IEEE Wireless Communications and Networking Conference (WCNC)*. IEEE, 2022.

Zheng, Licheng, Kim Khoa Nguyen, and Mohamed Cheriet. "Joint positioning and data communication provisioning using 5G pattern division multiple access (PDMA) airborne." *ICC 2024-IEEE International Conference on Communications*. IEEE, 2024.

Zheng, Licheng, Kim Khoa Nguyen, and Mohamed Cheriet. " Joint Airborne Wireless Positioning and Communication Services with Pattern Division Multiple Access (PDMA)." *IEEE Journal of Selected Area on Communication*. IEEE, 2026

Academic Peer Review

IEEE Transactions on Intelligent Transportation Systems (6 times), IEEE Transactions on Computers (3 times), Digital Communications and Networks (2 times)
IEEE International Conference on Communications (ICC, 2 times), IEEE Transactions on Engineering Management

Certificate

- Cisco Certificate Internetwork Expert (CCIE) Enterprise Infrastructure (Certificate #: 65896)
- Coursera Certificate Build Basic Generative Adversarial Networks (GANs)